Advanced Logic

Fuzzy Logic & Temporal Logic

Based on O'Regan – A Concise Guide to Formal Methods
Chapters 7.1-7.3

Advanced Topics in Logic

Fuzzy Logic

Mathematical model for vagueness

Degrees of truth (0 to 1)

Used in control systems and AI

Temporal Logic

Expresses time-dependent properties

LTL and CTL variations

Critical for model checking

Fuzzy Logic: Introduction

What is Fuzzy Logic?

Classical logic allows only **true (1)** or **false (0)** values. Fuzzy logic extends this to allow **degrees of truth** between 0 and 1.

Classical Logic

Is it hot outside?
Answer: Yes (1) or No (0)

Fuzzy Logic

Is it hot outside?
Answer: 0.7 (somewhat hot)

Why Fuzzy Logic?

Models Human Reasoning

Real-world concepts are often **vague and imprecise**. Fuzzy logic captures this vagueness mathematically, allowing systems to reason about uncertain or approximate information.

- **Vague terms:** *"tall", "warm", "expensive", "fast"*
- **Approximate descriptions:** *"close to the station", "moving fast"*
- **Gradual transitions:** Temperature changes from cold to warm to hot
- **Expert knowledge:** Rules like "if close and moving fast, brake harder"

Fuzzy Logic Applications



Control Systems

Train braking, air conditioning, washing machines



Artificial Intelligence

Expert systems, decision support, pattern recognition



Image Processing

Edge detection, image enhancement, compression



Natural Language

Linguistic variables, meaning representation

Fuzzy Sets & Membership

Traditional Set: An element either belongs (1) or doesn't belong (0) to a set.

Fuzzy Set: An element has a **degree of membership** ranging from 0 to 1.

Example: "Tall" People

Height 150cm: membership = 0.0 (not tall)
Height 170cm: membership = 0.3 (somewhat tall)
Height 180cm: membership = 0.7 (fairly tall)
Height 190cm: membership = 0.9 (very tall)
Height 200cm: membership = 1.0 (completely tall)

Temporal Logic

Reasoning About Time

What is Temporal Logic?

Temporal logic is a **modal logic** that expresses properties with **time dependencies**. It allows us to reason about how system properties change over time and specify requirements like "eventually" or "always".

Key Concepts

- **Time-based properties:** *"Request is eventually granted"*
- **Safety properties:** *"Bad state never happens"*
- **Liveness properties:** *"Good state eventually happens"*
- **Model checking:** Automatically verify temporal properties

Temporal Logic Operators

G (Globally)
Property holds at all future time points
Example: ***G(speed ≤ 100)***

F (Finally)
Property eventually holds at some future point
Example: ***F(task_complete)***

X (Next)
Property holds in the next time step
Example: ***X(door_open)***

U (Until)
Property φ holds until ψ becomes true
Example: ***request U grant***

Linear vs Branching Time

LTL

Linear Temporal Logic

- Linear time
- Single timeline
- Operators: G, F, X, U
- Intuitive for sequences

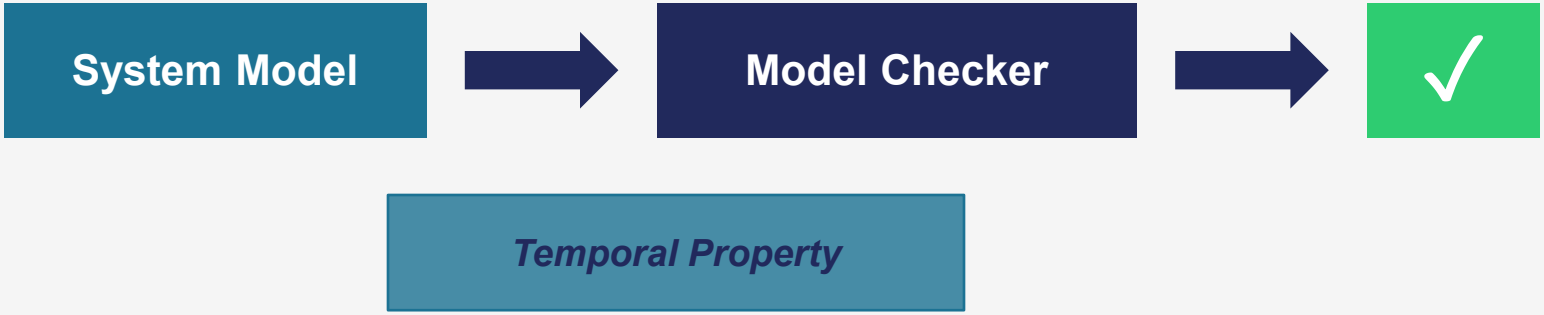
CTL

Computation Tree Logic

- Branching time
- Multiple futures
- Path quantifiers: A, E
- Faster model checking

Model Checking with Temporal Logic

Model checking is an **automated verification technique** that systematically checks whether a system model satisfies a temporal logic specification.



Process:

1. Model the system as a state machine
2. Specify properties in temporal logic (LTL/CTL)
3. Run model checker to verify all states
4. Get result: property holds or counterexample

Applications of Temporal Logic

1. **Hardware Verification**
   CPU design, memory controllers, circuit correctness

2. **Protocol Verification**
   Network protocols, communication systems, distributed algorithms

3. **Software Verification**
   Concurrent programs, operating systems, safety-critical software

4. **Aerospace Systems**
   Air traffic control, autopilot systems, flight control

Fuzzy Logic vs Temporal Logic

Fuzzy Logic

Focus: Vagueness

Deals with: Uncertainty in values

Used for: Control, AI, decision-making

Temporal Logic

Focus: Time dependencies

Deals with: Properties over time

Used for: Verification, model checking

Key Takeaways

- ✓ Fuzzy logic extends classical logic with degrees of truth
- ✓ Models vagueness and imprecision in real-world concepts
- ✓ Temporal logic expresses time-dependent properties
- ✓ LTL and CTL are key variants for model checking
- ✓ Both are essential tools in formal methods

Thank You

Questions?